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THE ROTATION OF SPAIN: PALAEOMAGNETIC EVIDENCE FROM THE EASTERN PYRENEES

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SUMMARY

The results of a palaeomagnetic investigation on igneous and sedimentary Permo-Triassic rocks from the Spanish Pyrenees are presented and some conclusions are drawn. During demagnetization with a.c. magnetic fields and after correction for the dip, 41 andesitic samples revealed a characteristic magnetization with an average declination (D) of 169.5° and an inclination (I) of −3° for the Lower Permian. The more tentative results of 17 pelitic samples are given as three site means with $D = 162°$, $I = -3.5°$ and $D = 152.5°$; $I = -33°$ for the Permian undifferentiated and $D = 340.5°$, $I = +24°$ for the Lower Triassic.

The virtual pole position for the Lower Permian andesites is at 48.5° N 163° W. This virtual pole position, deviating from already known data for the Permian of stable Europe, can be best explained by a post-Permian counterclockwise rotation of Spain in the order of 30°.

INTRODUCTION

As part of a palaeomagnetic investigation program carried out by members of the State University of Utrecht, 94 samples from the Sierra del Cadi (Huesca province, Spain) were taken in the summer of 1966 in order to obtain paleomagnetic data from the eastern Pyrenees. Some years ago similar investigations in the Spanish Pyrenees by VAN DER LINGEN (1960) and SCHWARZ (1962, 1963) yielded virtual Permo-Triassic pole positions that were different from those for stable (extra alpine) Europe. Their results indicated a counterclockwise rotation of Spain (GIRDLER, 1965), postulated previously by several authors (DU TOIT, 1937; CAREY, 1958; BULLARD et al., 1965). Recent investigations (VAN DER VOO, 1967, and the present paper) on volcanic and sedimentary rocks lead to firmer based palaeomagnetic evidence for the rotation of the Iberian peninsula.

GEOLOGICAL SITUATION

The region from which the andesitic and sedimentary samples for the present study were taken is situated near Seo de Urgel, some 20 km south of Andorra (Fig.1).

Between the Hercynian and the Alpine orogenies, two periods of volcanic activity occurred in the Pyrenees. The first took place in the Lower Permian and the second in the Lower Triassic. The former resulted mainly in effusives near the Pic du Midi d'Ossau and near the Sierra del Cadi, whereas the latter resulted in many lava beds along the north flank of the Pyrenees. In the Sierra del Cadi we find the Lower Permian volcanic rocks mainly exposed in the foothills on the north side (Fig.2), upon which more or less conformable the Permo-Triassic, the Upper Triassic, the Senonian, the Garumnian and the Eocene together form the range of the Cadi. The contact with the underlying Paleozoic is in general badly exposed. At some places, however, it was posssible to observe that the lava beds were lying conformably upon Stephanian schists. For an extensive description of this region the reader is referred to BOISSEVAIN (1934). A recent petrographical and petrochemical study on the volcanic rocks was carried out by MORRE and THIÉBAUT (1964). They found that the lavas had a composition of hyalodacites and hyaloandesites, which had been partly and at varying degree been altered by means of a spilitisation into albitophyries. This alteration seems to have been contemporaneous with the deposition of the lava beds (MORRE and THIÉBAUT, 1964).

The greater part of the volcanic series, however, consists of tuffs and tuff-breccia. The dip of the volcanic beds is generally south and varies between 10° and 60°.

The 54 andesitic samples were taken from 13 different sites (Fig.2) that represent several flows. It was not always possible to correlate the flows of one locality with those of another and thus no exact number of lava flows can be given.

The pelitic samples were collected from three sites and all from different beds:

(*1*) The first group of six samples was taken from the centre of the area directly resting upon the andesites; the age of this group, however, is difficult to determine as there appeared to be a stratigraphical hiatus between the lavas and the sediments.

(*2*) A second group of seven samples was taken south of Seo de Urgel in the valley of the Sègre, on the eastern valley-side, in the lower part of the series.

(*3*) The third group of four samples was also taken south of Seo de Urgel, but on the western valley-side much higher in the series, probably already in the Lower Triassic (Fig.2).

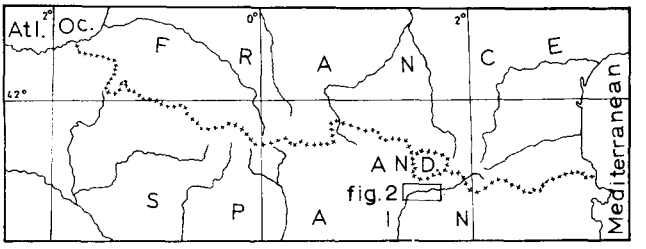


Fig.1. Location of the palaeomagnetic sampling area. The outlined area is given in detail in Fig.2.

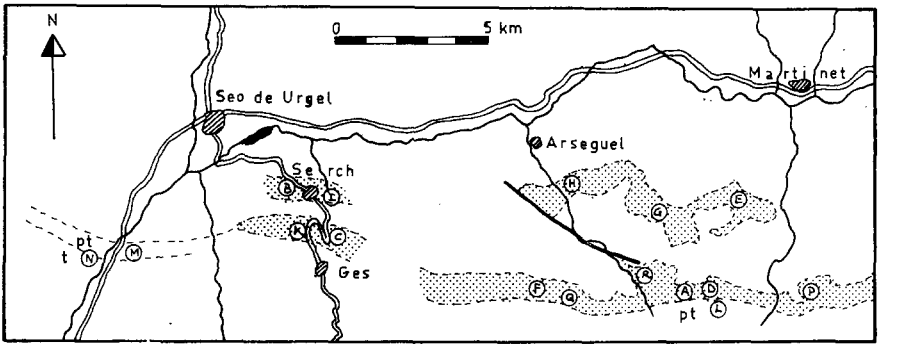


Fig.2. Schematic geological map of the Lower Permian outcrops (shaded) and of some Permo-Triassic exposures (pt) in which the sampling sites (*A–R*) are situated. (After BOISSEVAIN 1934.)

METHODS OF RESEARCH

The samples were embedded with their correct orientation in paraffin cubes of 10 cm edges, after having been sawn to approximately cubic shape in order to minimize magnetic shape effects. The directions and intensities of the N.R.M. (natural remanent magnetization) were measured in the Paleomagnetic Laboratory of the Utrecht University by means of astatic magnetometers. Thereafter all samples were treated with a progressive demagnetization with a.c. magnetic fields up to 3,000 Oe (peak value), in order to obtain an analysis of the magnetic components. For an extensive description of the methods used for measurement and demagnetization the reader is referred to As and ZIJDERVELD (1958); ZIJDERVELD (1967a). The data thus obtained are summarized in Table I and plotted in stereograms (Fig.3 and 6).

RESULTS OF THE LOWER PERMIAN ANDESITIC SAMPLES (SAMPLES CAD 6–78)

From thirteen sites 54 samples were taken. The intensity of their N.R.M.

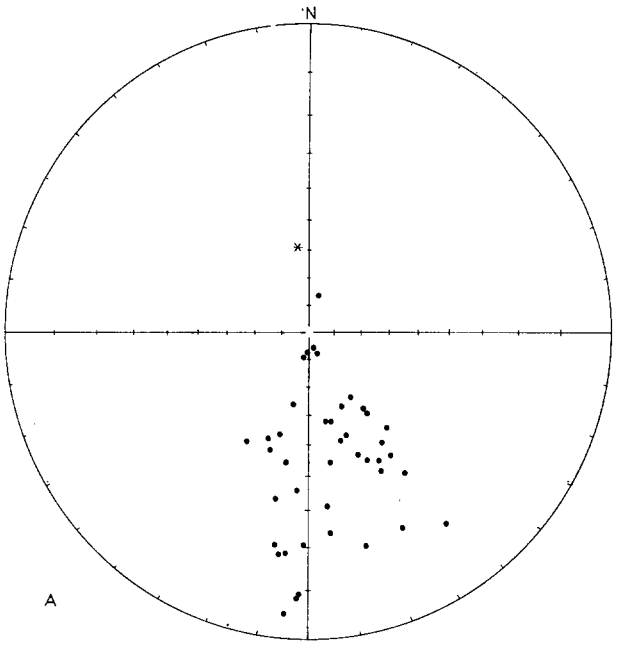
TABLE I

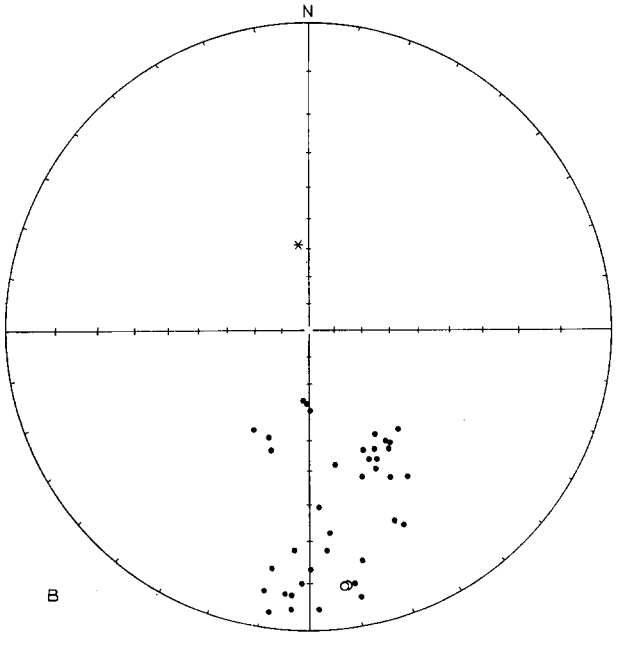
SUMMARY OF MEASUREMENTS

Site	*Sample No.*	*N.R.M.*				*Characteristic component*		*Characteristic component after tectonic correction*	
		D(°)	*I(°)*	*Joules/ cm³*	*Q*	*D(°)*	*I(°)*	*D(°)*	*I(°)*
Quer	CAD 6	151	36	8.54	1.2	151	32	155.5	— 4
Foradat	CAD 7	144.5	31.5	7.42	1.1	146	30	151	— 4.5
(*A*)	CAD 8	173	30	5.19	0.9	176.5	29.5	177	−10.5
Bastanist	CAD 9	300	63	93.85	11.0	300	63		
(*P*)	CAD 10	207	64.5	101.06	10.1	207	64.5		
	CAD 11	155	71	14.58	2.3	157	70.5		
	CAD 12	145	−49	15.44	1.1	142.5	−44.5		
Serch (*B*)	CAD 13	184.5	5	429.51	17.4	184	4.5	184	−16
	CAD 14	181.5	9	270.76	2.9	182	10	182	−10.5
	CAD 16	182.5	8.5	166.29	1.9	184	7.5	184	−12.5
	CAD 17	173	23.5	874.40	40.2	174.5	22.5	174	3
	CAD 18	186	18.5	132.17	6.2	184	18	184	— 3
	CAD 19	188.5	20	196.92	2.1	190	8	190	−13
	CAD 20	187.5	18	178.20	1.9	188.5	3.5	189	−17
Serch	CAD 22	190.5	80.5	83.59	1.9	183.5	64	169.5	4
(*C*)	CAD 23	156	80	59.03	1.7	179	60.5	168	2
	CAD 24	182	82.5	18.0	0.9	183	63.5	169	3.5
Estana	CAD 28	143	15	265.0	186.0	157	21	160.5	— 8.5
(*D*)	CAD 29	154.5	40.5	161.58	49.1	155	40	173	6.5
	CAD 31	153	19.5	28.77	3.5	154	19	158	— 9
S. Mosbé	CAD 37	157.5	43.5	459.82	1.8	154	36.5	161.5	— 7.5
(*E*)	CAD 38	159	50.5	427.09	1.8	156.5	43.5	166	— 2
Ansovell	CAD 39	33.5	−63.5	327.05	7.3	27.5	−62.5		
(*Q*)	CAD 40	33	45	252.14	6.7	32.5	46.5		
	CAD 41	19	−61.5	312.24	8.6	19	−72		
	CAD 42	30.5	−63	327.15	10.3	20.5	−63		
Ansovell	CAD 44	145	43.5	9.36	1.0	146	40	159	— 7.5
(*F*)	CAD 45	139.5	46	33.83	4.6	137.5	43	162.5	— 1.0
	CAD 46	144	56	35.71	2.1	147	46	163.5	— 3
	CAD 47	144	54	62.35	3.0	145	42.5	160	— 5
Barbuja	CAD 49	146	61.5	199.31	1.1	152	39	161.5	3.5
(*G*)	CAD 50	162.5	49.5	235.75	1.2	160	36	164.5	— 1
	CAD 51	155	60	116.41	1.1	160	36	164.5	— 1

TABLE I (*continued*)

Site	*Sample No.*	*N.R.M.*				*Characteristic component*		*Characteristic component after tectonic correction*	
		D(°)	*I*(°)	*Joules/cm*^{3}	*Q*	*D*(°)	*I*(°)	*D*(°)	*I*(°)
Quer	CAD 54	203	52	332.87	24.0	202.5	48.5		
Foradat	CAD 55	296.5	46.5	68.71	7.1	295.5	44.5		
(*R*)	CAD 56	245	− 7.5	221.67	9.9	243	− 7		
	CAD 57	177	−14.5	109.52	10.3	183	− 5		
	CAD 58	305	38.5	77.53	7.4	305	38		
Arseguell	CAD 60	145	38.5	156.05	17.7	144	41.5	157	1.5
(*H*)	CAD 61	169.5	43.5	21.41	1.8	169	41.5	174	− 7
	CAD 62	150.5	39	141.83	16.55	153	42	164	− 1.5
Serch	CAD 66	209	45.5	8.77	1.0	209	49	189	0.5
(*I*)	CAD 67	200	49.5	9.82	1.0	200.5	48.5	184.5	− 3.5
	CAD 68	197.5	46	10.32	1.1	197.5	44.5	185.5	− 8
Serch-Gès	CAD 69	159	84	48.38	0.3	172.5	− 9	173	−24
(*K*)	CAD 70	191	63	58.40	0.4	178	13.5	177.5	− 1.5
	CAD 71	189.5	43.5	48.52	0.4	169	6	169	− 9.5
	CAD 72	17	76	85.59	0.4	185.5	7.5	185.5	− 6.5
	CAD 73	183.5	35.5	57.24	0.4	178	4.5	178.5	−10.5
	CAD 74	165	56.5	88.65	0.3	170	9.5	170	− 6
	CAD 75	181	20.5	24.85	0.3	171.5	− 9	172	−24
	CAD 76	191	32.5	123.05	0.6	185.5	7.5	185.5	− 6.5
	CAD 77	195	52	138.09	0.7	189	13	188.5	− 1
	CAD 78	168.5	57	149.32	0.8	175.5	18	175	3
Estana	CAD 27	187	73.5	9.48	1.4	173.5	49.5	177	−32.5
(*L*)	CAD 79	76	69	16.94	1.8	124.5	56	146.5	−26
	CAD 80	91	72.5	20.98	1.9	133	46	141.5	−34.5
	CAD 81	90.5	70.5	22.47	1.7	131.5	46.5	141.0	−33.5
	CAD 82	160	60	22.26	1.7	160.5	46.5	161.5	−43.5
	CAD 83	94.5	74.5	15.79	1.6	133	57	153	−23.5
Seo de	CAD 84	150	53.5	4.12	0.6	151	36.5	160.5	0.5
Urgel (*M*)	CAD 85	147	52.5	8.39	1.8	149.5	37.5	160.5	3
	CAD 86	153.5	47	5.69	1.5	155	33.5	162.5	− 3.5
	CAD 87	158	28	12.43	2.4	158	20.5	165	−16.5
	CAD 88	150	62.5	8.46	1.4	149.5	46.5	168	− 1
	CAD 89	60.5	70	2.53	0.8	99.5	63		
	CAD 90	8	37.5	15.08	0.7	122	52		
Seo de	CAD 91	332.5	18.5	3.71	0.5	344.5	−20	344	19.5
Urgel (*N*)	CAD 92	332.5	32.5	6.61	0.9	341.5	− 9	335	28
	CAD 93	337	− 4.5	9.22	1.0	336	−23.5	340	12.5
	CAD 94	353	19.5	5.11	0.7	350.5	− 7.5	342.5	34


N
A


N
B

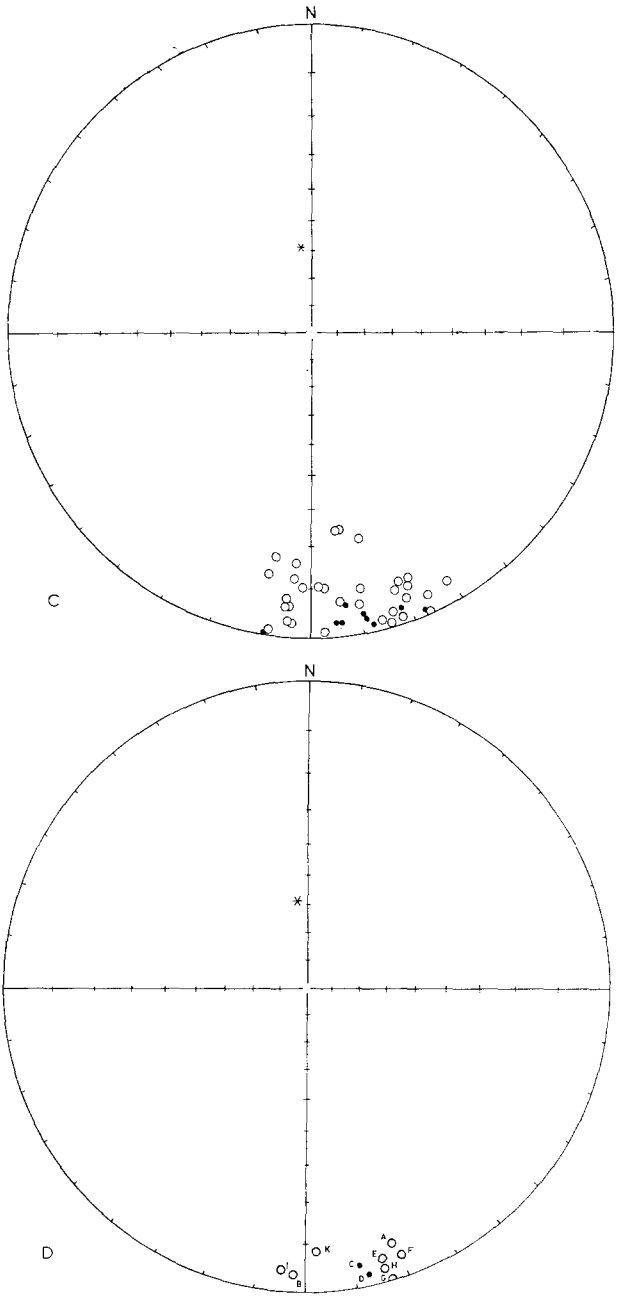


Fig.3. Stereographic projection of the magnetization direction of the Lower Permian andesitic samples: A. Total natural remanent magnetization without tectonic correction; B. Characteristic directions revealed by partial demagnetization, without tectonic correction; C. Same as B, corrected for the geological dip; D. Site means *A–K* (Table II), computed from the characteristic directions after tectonic correction.

Full circles denote north seeking directions pointing downward and open circles denote north seeking directions pointing upward. The asterisk denotes the local present-day geomagnetic field direction.

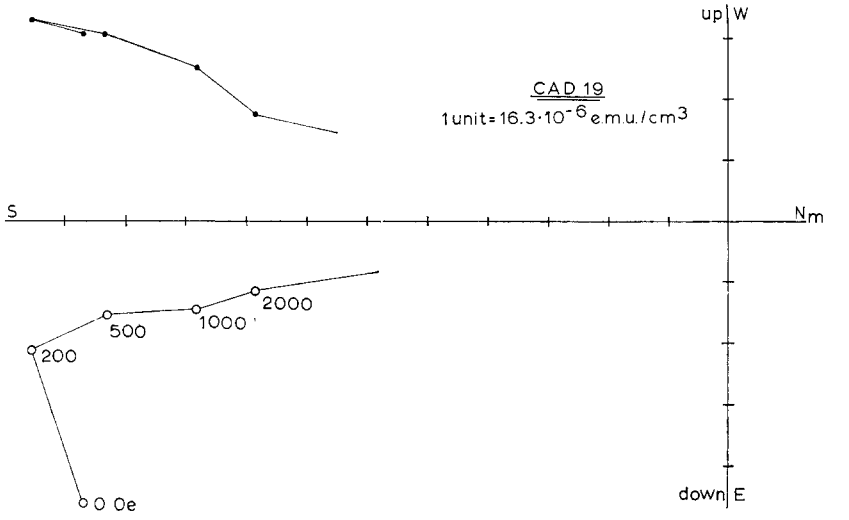


Fig.4. Demagnetization diagram of the andesitic sample CAD 19. Plotted points represent successive positions—in orthogonal projection—of the end of the magnetic vector during progressive demagnetization. Full symbols represent projections on the horizontal plane; open symbols projections on the north-south vertical plane.

Numbers denote a.c. field intensities in Oersted and N_m denotes the magnetic north.

TABLE II

SITE MEANS[1]

Site	*Samples*	*Age*	*Dip correction*	*Mean characteristic direction*			
				a.c. field range	*D(°)*	*I(°)*	$\alpha_{95}(°)$
A. Quer Foradat	CAD 6–8	L.P.	90–40	400–2,000	161	−7.5	22
B. Serch	CAD 13–20	L.P.	95–20	500–3,000	183.5	−4	6
C. Serch	CAD 22–24	L.P.	70–60	400–2,000	169	4.5	4
D. Estana	CAD 28, 29, 31	L.P.	135–50	200–1,000	167	2.5	18
E. S. Mosbé	CAD 37, 38	L.P.	100–50	500–2,000	164	−4.5	—
F. Ansovell	CAD 44–47	L.P.	105–60	500–1,000	160	−4.5	5
G. Barbuja	CAD 49–51	L.P.	95–40	1,000–2,000	163	0	6
H. Arseguell	CAD 60–62	L.P.	100–50	400–3,000	164	−3	15
I. Serch	CAD 66–68	L.P.	70–60	200–2,000	185.5	−4.5	7
K. Serch-Gès	CAD 69–78	L.P.	75–15	500–3,000	177.5	− 9	7
L. Estana	CAD 27, 79–83	P.	85–95	1,000–3,000	152.5	−33	10
M. Seo de Urgel	CAD 84–88	P.	105–50	1,500–3,000	162	−3.5	8
N. Seo de Urgel	CAD 91–94	L.Tr.	115–50	3,000–	340.5	+24	11

[1]Letters A–K, see fig.2 and 3D; L.P.: Lower Permian; P. = Permian; L.Tr.: Lower Triassic. Azimuths corrected for the local present day geomagnetic declination (6° W).

varied from 5 to $875 \cdot 10^{-6}$ e.m.u./cm^3 and the Q-value from 0.3 to 186. The directions of the N.R.M.'s are given in Fig.3A. During demagnetization all the samples lost a secondary component between the a.c. magnetic field values of 0 and 500 Oer (peak value) after which the decrease of the magnetization vector went in a direction that passed more or less through the origin of the co-ordinate system. As the first eliminated component was conformable to the present-day geomagnetic field direction it was assumed to be a secondary component (Fig.4, 5). The remaining components, after correction for the geological dip, appeared to have a quite similar direction, which should therefore be the characteristic one (Fig.3C).

From three sites the results were not used for computation (Table I):

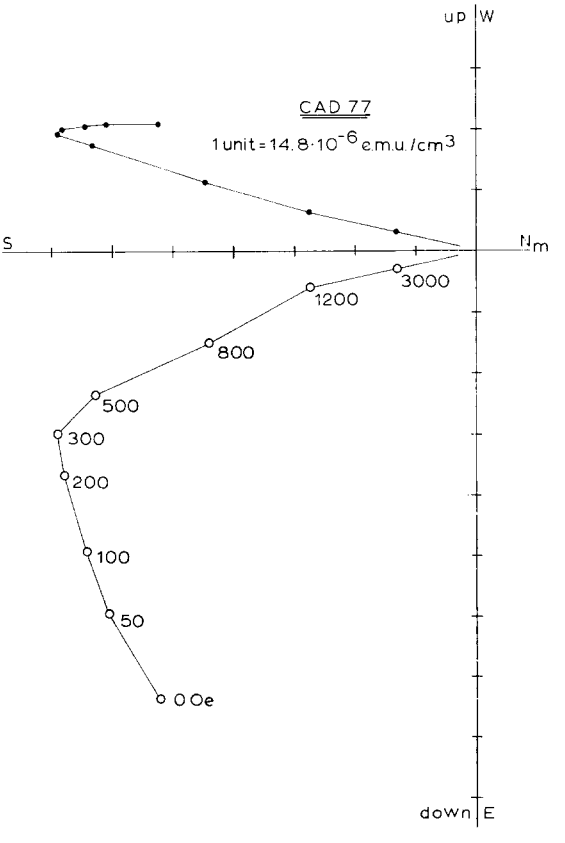
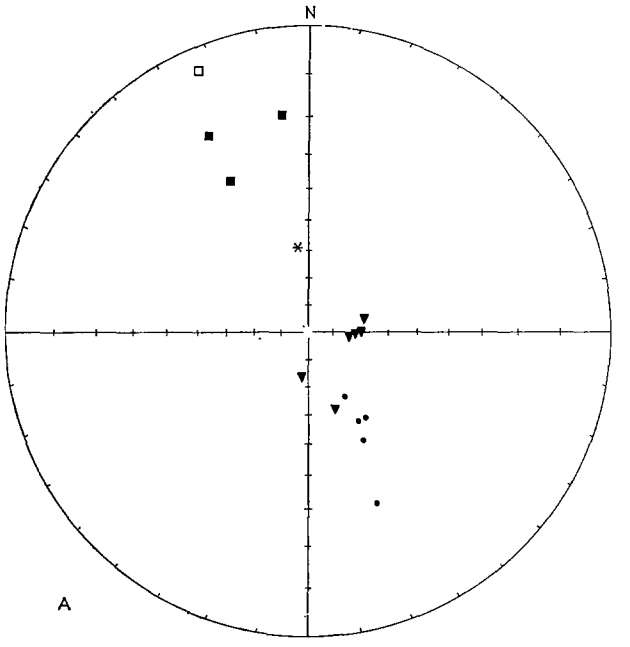
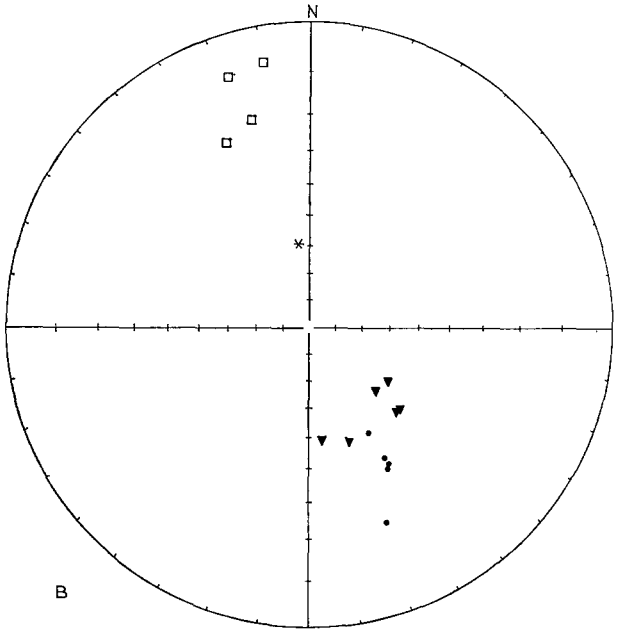


Fig.5. Demagnetization diagram of the andesitic sample CAD 77. Plotted points represent successive positions—in orthogonal projection—of the end of the magnetic vector during progressive demagnetization. Full symbols represent projections on the horizontal plane; open symbols projections on the north–south vertical plane. The numbers denote the a.c. field intensities in Oe and N_m denotes the magnetic north.


N
A


N
B

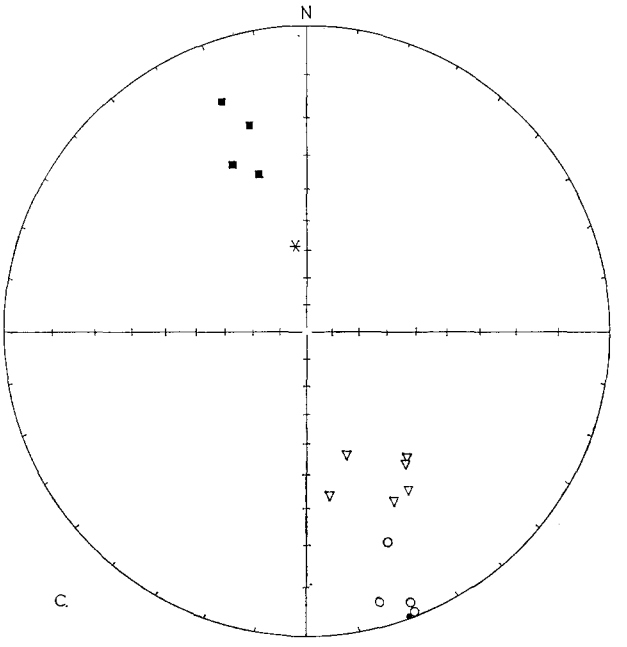


Fig.6. Stereographic projection of the magnetization directions of the Permo-Triassic sediments: A. Total natural remanent magnetization without tectonic correction; B. Characteristic directions revealed by partial demagnetization, without tectonic correction; C. Same as B, corrected for the geological dip.

Full and open circles denote series CAD 84–88, triangles denote series CAD 27, 79–83 and squares denote series CAD 91–94. Full symbols denote north seeking directions pointing downward and open symbols denote north seeking directions pointing upward. The asterisk denotes the local present-day geomagnetic field direction.

the results from the Bastanist site (*P*) were omitted because the dip of the bedding could not be determined. The samples from the Ansovell I-site (*Q*) were taken from a block lava, which may have moved after solidification. The samples from the Quer Foradat II-site (*R*) showed such a wide scatter of directions, although they were taken from a relative small exposure, that they had to be left out.

Ten site means have been calculated (Table II and Fig.3D) from which the average magnetization direction for this region for the Lower Permian has been computed, giving unit weight to each site mean: $D = 169.5°$, $I = -3°$, $\alpha_{95} = 6°$. When giving unit weight to each sample, the semiangle of the cone of 95% confidence is 4°.

RESULTS OF THE PERMO-TRIASSIC PELITES

As the three groups of pelites could not be correlated with each other, the results will be given separately.

The first series (CAD 27, 79–83) had N.R.M. intensities that varied between 9.5 and 22.5 · 10^{-6} e.m.u./cm^3 and Q-value between 1.4 and 1.9. Between a.c. magnetic field values up to 1,000 and 2,000 Oe a secondary component was eliminated, whereupon at higher values the characteristic magnetization was revealed (Fig.6—triangles). The average of this series is: $D = 152.5°$, $I = -33°$ and $\alpha_{95} = 10°$.

The second series (CAD 84–90), taken south of Seo de Urgel, had somewhat lower intensities, from 2.5 to 15 · 10^{-6} e.m.u./cm^3 and Q-values between 0.6 and 2.4. These sedimentary samples reacted quite similar as the preceding series to a.c. demagnetization (Fig.7). The average direction from this series is (CAD 89 and 90 not included because they could not be sufficiently cleaned): $D = 162°$, $I = -3.5°$ and $\alpha_{95} = 8°$ (Fig.6—circles and dots).

The third and probably Lower Triassic series of only four samples (CAD 91–94) had N.R.M. intensities that varied between 3.7 and 9.2 · 10^{-6} e.m.u./cm^3 and Q-values between 0.5 and 1.0 and a normal N.R.M. direction. These samples showed a stronger change, especially in the inclination during demagnetization (Fig.8), from which it is not quite clear whether the secondary component had been completely removed. They gave the following average direction: $D = 340.5°$, $I = +24°$ and $\alpha_{95} = 11°$ (Fig.6—squares).

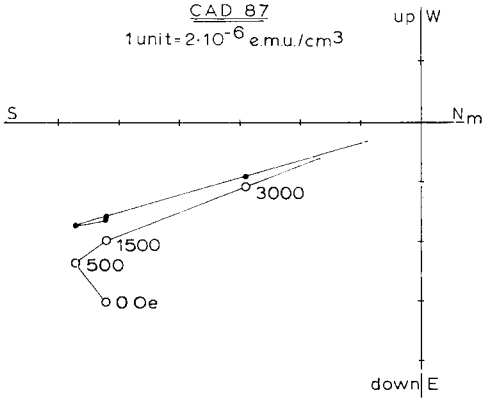


Fig.7. Demagnetization diagram of sediment sample CAD 87 (Permian undifferentiated). Plotted points represent successive positions—in orthogonal projection—of the end of the magnetic vector during progressive demagnetization. Full symbols represent projections on the horizontal plane, open symbols projections on the north-south vertical plane.

The numbers denote a.c. field intensities in Oe and N_m denotes the magnetic north.

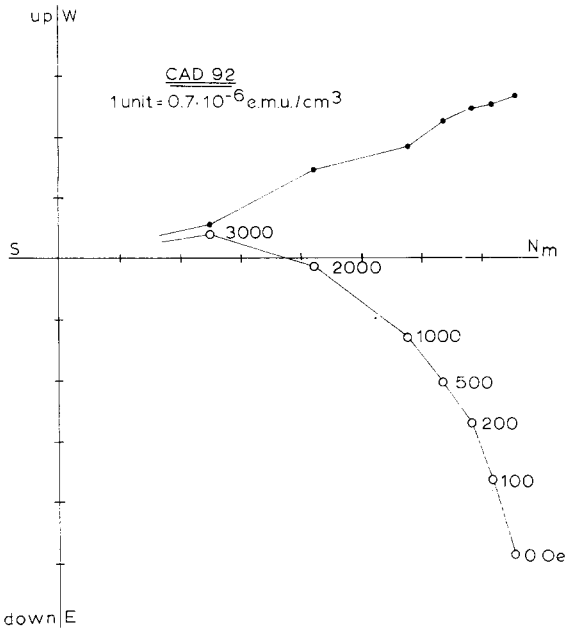


Fig.8. Demagnetization diagram of the sediment sample CAD 92 (Lower Triassic). Plotted points represent successive positions—in orthogonal projection—of the end of the magnetic vector during progressive demagnetization. Full symbols represent projections on the horizontal plane; open symbols projections on the north–south vertical plane.
The numbers denote the a.c. field intensities in Oe and N_m denotes the magnetic north.

CONCLUSIONS

The computed declinations and inclinations gave the following pole positions (see also Table III):

Lower Permian	48.5° N 163°W (CAD 6–78, 41 samples, 10 sites)
Permian (undiff.)	47° N 154°W (CAD 84–88, 7 samples, 1 site)
Permian (undiff.)	56.5° N 128°W (CAD 27, 79–83, 6 samples, 1 site)
Lower Triassic	54.5° N 142°W (CAD 91–94, 4 samples, 1 site)

These data, of which only the first comprises a sufficient number of samples to be useful for further conclusions, deviate from other known Permian pole positions of stable Europe. In Table III the most reliable paleomagnetic results from Permian rocks of Europe—which were analysed by means of thermal, a.c. demagnetization or other stability tests—are given and it is quite clear that there exists a divergence of 25°–30° in declination between the stable Europe directions and the here presented Lower Permian direction. On the other hand, the inclination (−3°) fits very well in the map of the Permian isoclines (Fig.9), derived from the data of stable Europe summarized in Table III. The obvious conclusion is a

counterclockwise rotation of Spain of about 30°. (As there are up until now too few reliable paleomagnetic data from stable Europe as well as from Spain available, it would be premature to give the amount of rotation with greater accuracy).

The rotation of Spain has been suggested by several authors (DU TOIT, 1937; CLEGG et al., 1957; CAREY, 1958; BULLARD et al., 1965 and on paleomagnetic indications by GIRDLER, 1965). Together with the Silurian and Triassic paleomagnetic results from VAN DER VOO (1967), that are presented in this same issue, the Lower Permian direction described here gives strong paleomagnetic evidence for the supposed rotation.

The paleomagnetic directions derived from the sedimentary samples (CAD 27, 79–94), although not very reliable because of the low number of samples available, support this rotation.

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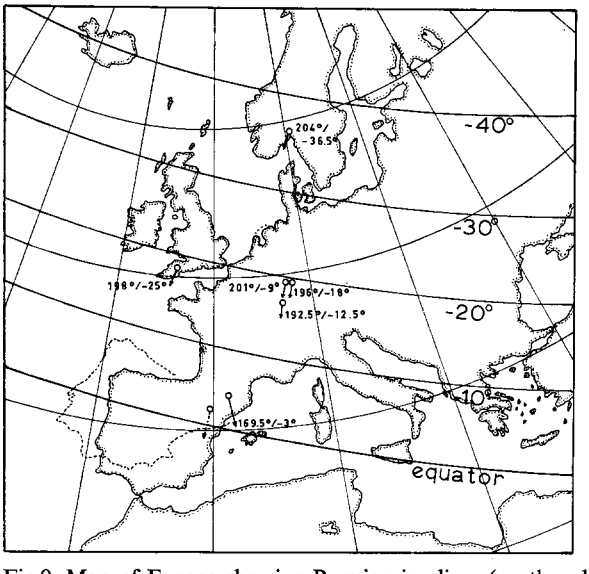


Fig.9. Map of Europe showing Permian isoclines (north pole at 47.7°N 160.5°E) derived from measurements on rocks from stable parts of the European continent (after ZIJDERVELD, 1967b). The data used to compose this map are listed in Table III, *1–4*. For comparison Spain has been rotated till the declination and inclination of the Permian of the Sierra del Cadi is in accordance with the isoclines, after which the Iberian peninsula occupies the position indicated by dotted lines. Accordance is achieved after a rotation of 27°.

TABLE III

SOME PERMIAN DIRECTIONS OF MAGNETIZATION AND ANCIENT POLE POSITIONS IN STABLE EUROPE AND SPAIN

Locality	*Type of rock*	*Characteristic magnetization*		*Number of samples*	*pole position*	α_{95}
		D(°)	*I*(°)	*N*		
1. Oslo region (Norway)	*I*	204	−36.5°	484	47° N 157° E	3°
2. Nahe region (a) (Germany)	*I*	201	− 9°	28	42° N 163° E	4°
Nahe region (b) (Germany)	*I*	196	−18°	34	48° N 168° E	8°
3. Nideck region (France)	*I*	192.5	−12.5°	37	47° N 169° E	5°
4. Exeter (England)	*I*	198	−25°	23	49,5°N 148,5°E	7°
5. Serra del Cadi (Spain)	*I*	169.5	− 3°	41	48,5°N 163° W	6°
6. Aragón Subordán (Spain)	*I*	152	−22.5°	14	51° N 133° W	6°
7. Anayet andesites + pelites (Spain)	*I* + *S*	164	−14°	11	52° N 154° W	10°

1 = VAN EVERDINGEN (1960), *2* = NIJENHUIS (1961), *3* = ROCHE et al. (1962), *4* = ZIJDERVELD (1967b), *5* = VAN DONGEN (1967), *6* = SCHWARZ (1962, 1963), *7* = VAN DER LINGEN (1960).
Type of rock: *I* = igneous, *S* = sediment; α_{95}: semiangle of 95% confidence after Fisher; *D* and *I* = declination and inclination.

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